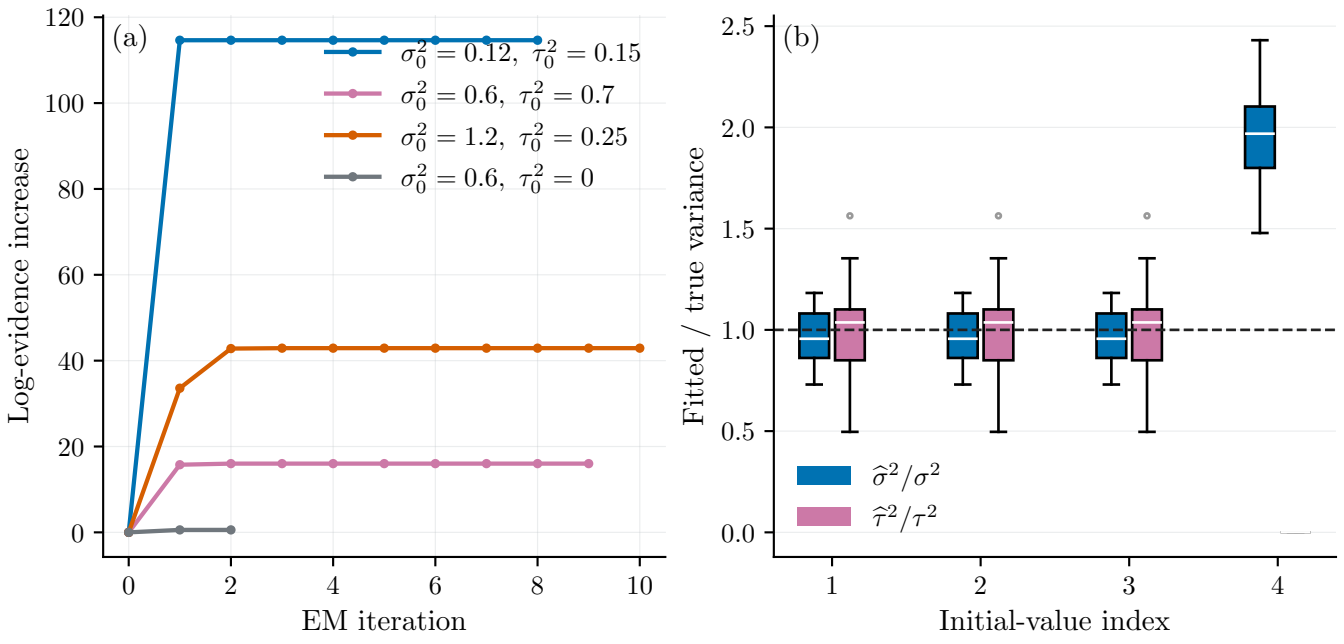
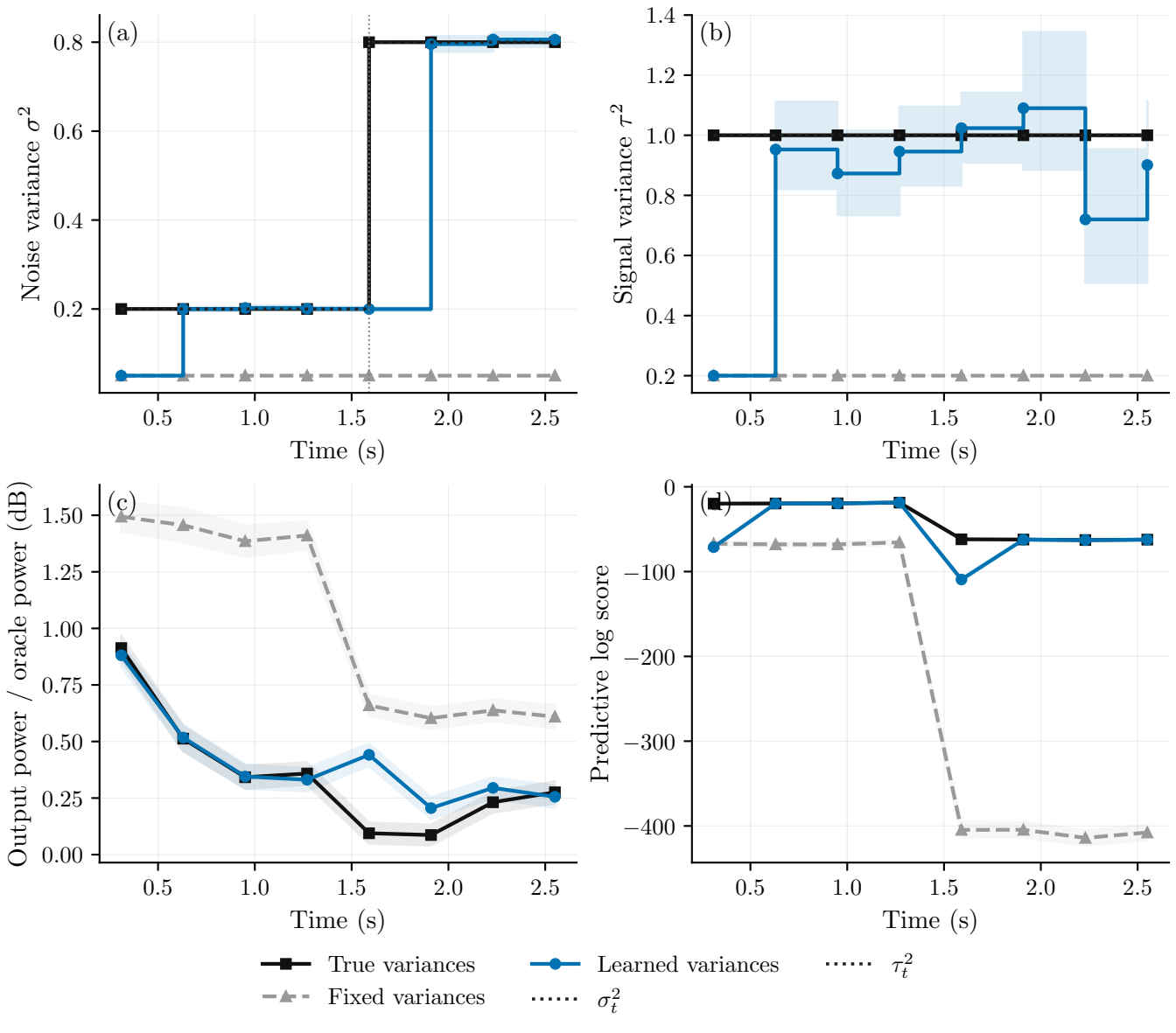
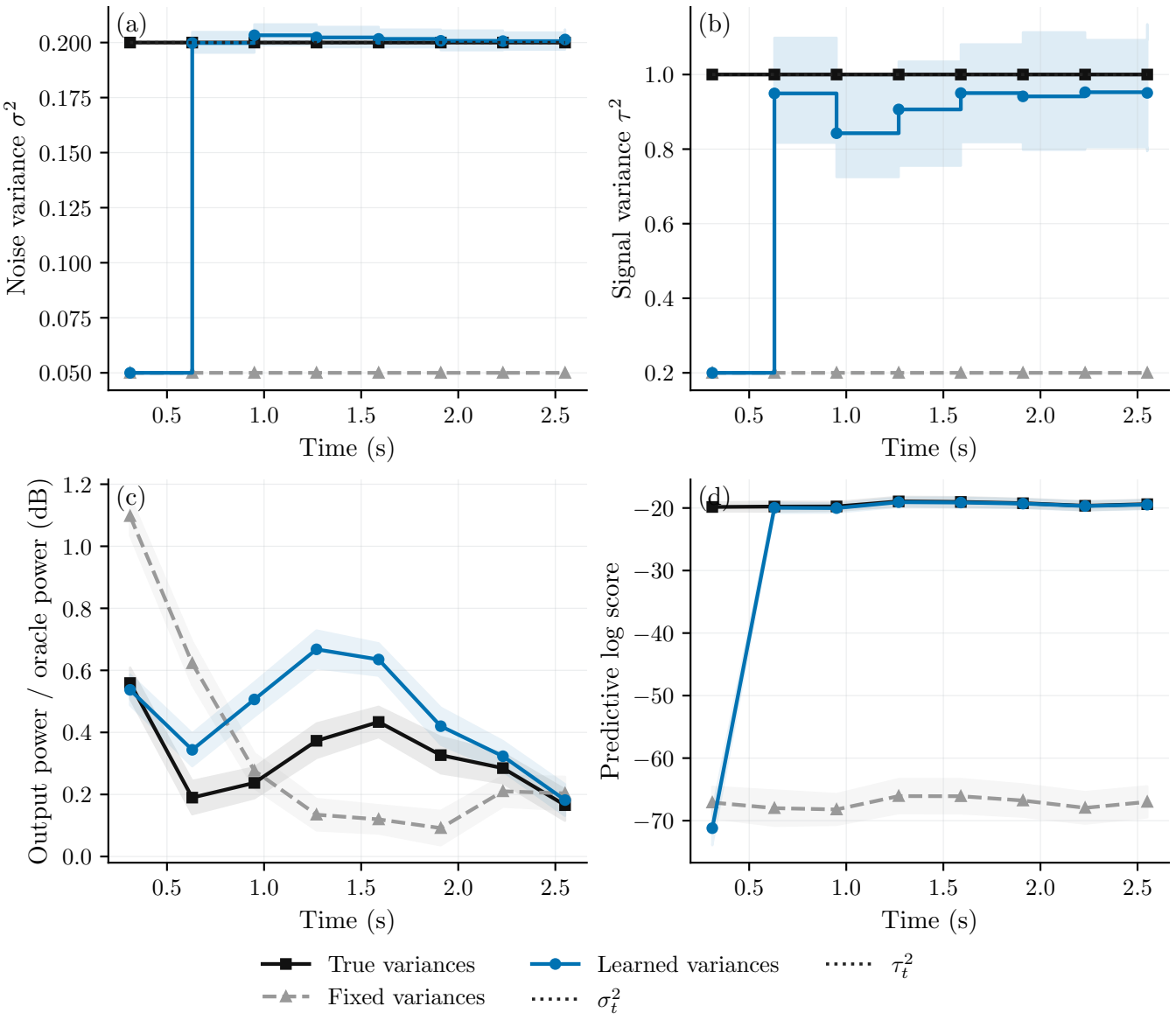
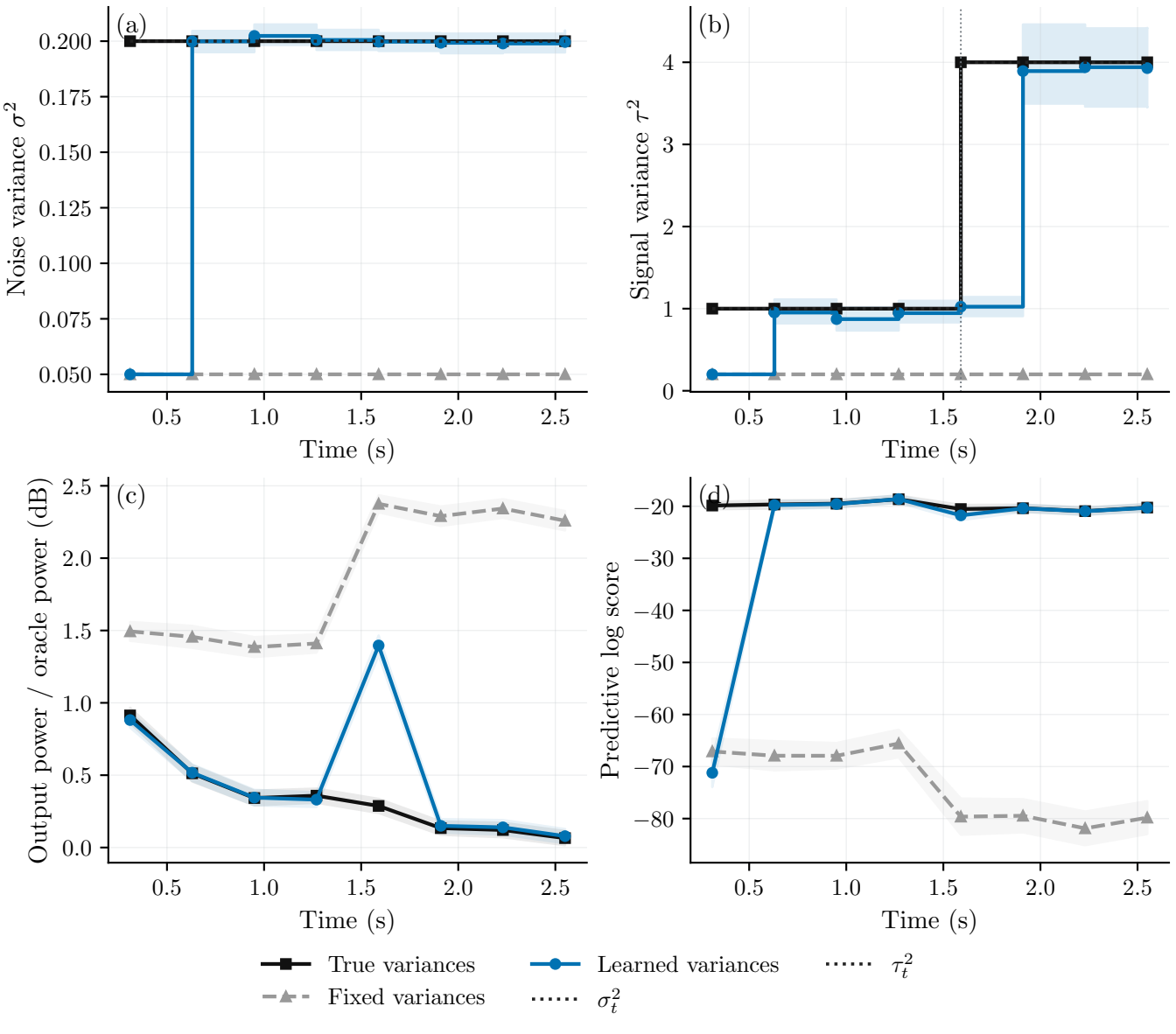
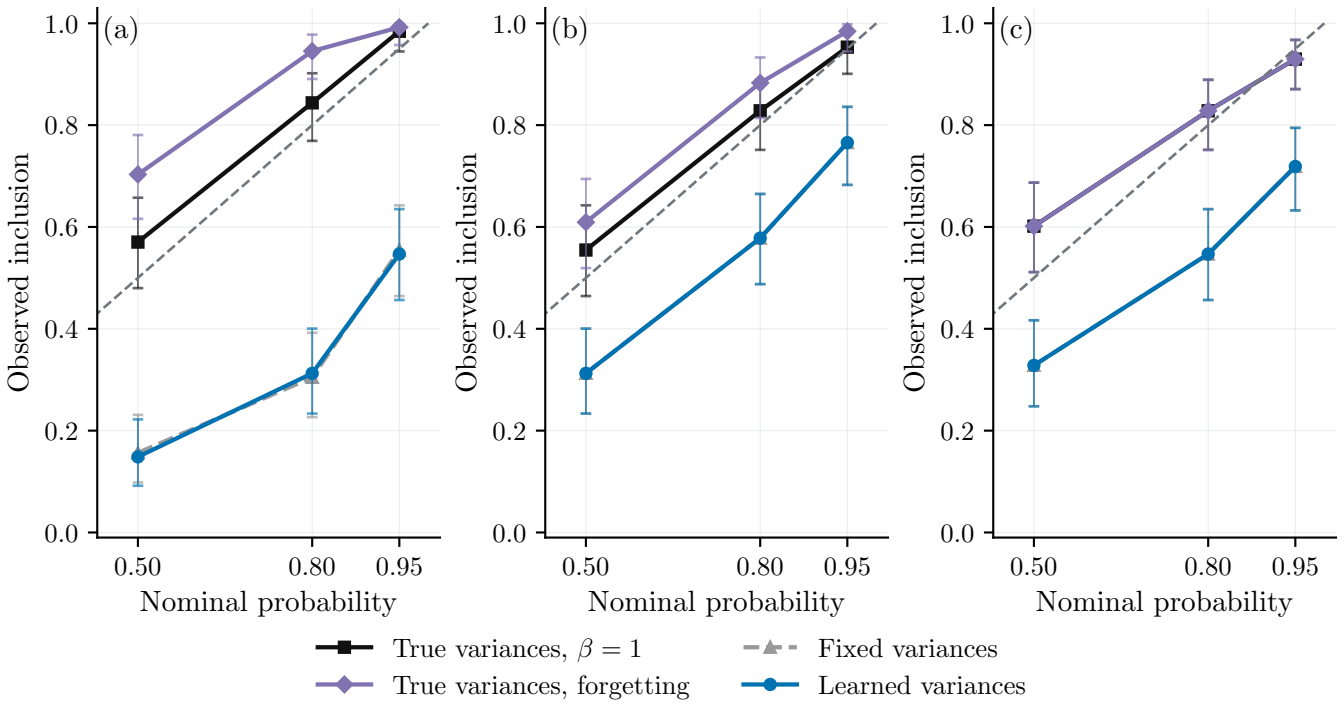
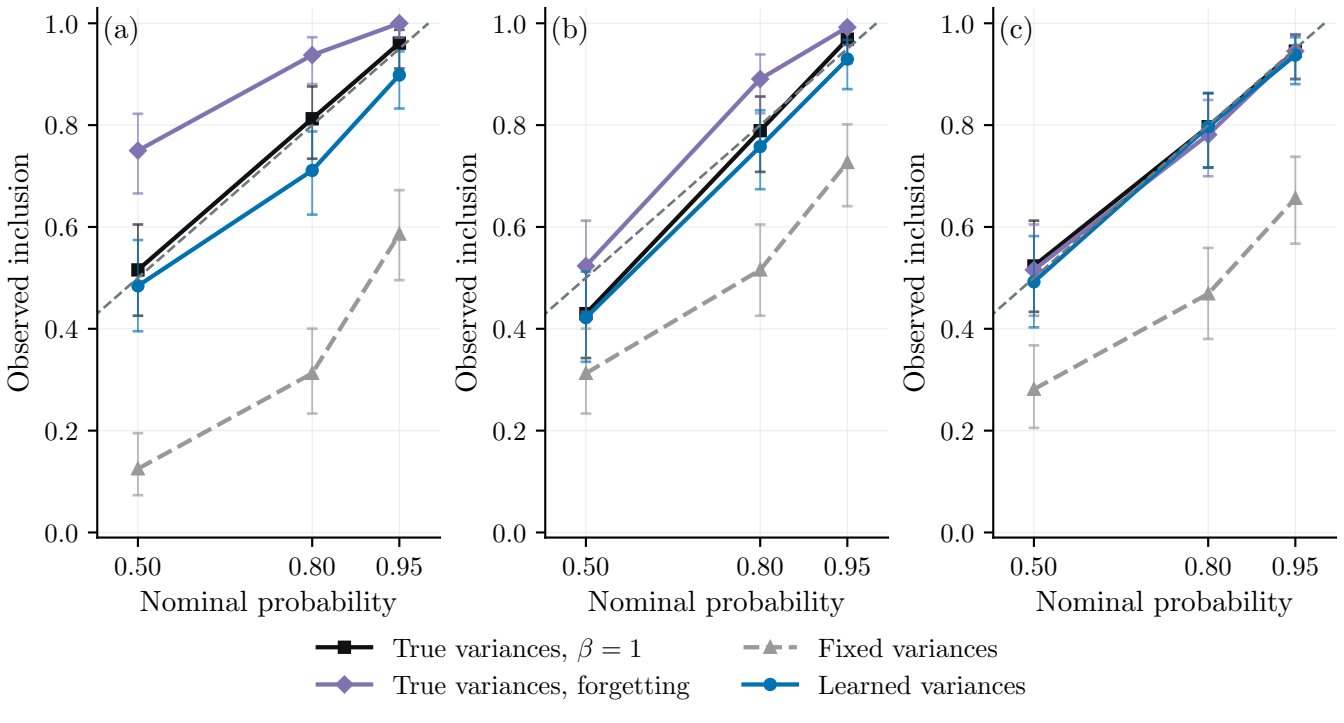

(a)
$\sigma_0^2 = 0.12,\ \tau_0^2 = 0.15$
$\sigma_0^2 = 0.6,\ \tau_0^2 = 0.7$
$\sigma_0^2 = 1.2,\ \tau_0^2 = 0.25$
$\sigma_0^2 = 0.6,\ \tau_0^2 = 0$
Log-evidence increase
EM iteration
0
20
40
60
80
100
120
0
2
4
6
8
10
(b)
Fitted / true variance
Initial-value index
$\widehat{\sigma}^2/\sigma^2$
$\widehat{\tau}^2/\tau^2$
0.0
0.5
1.0
1.5
2.0
2.5
1
2
3
4

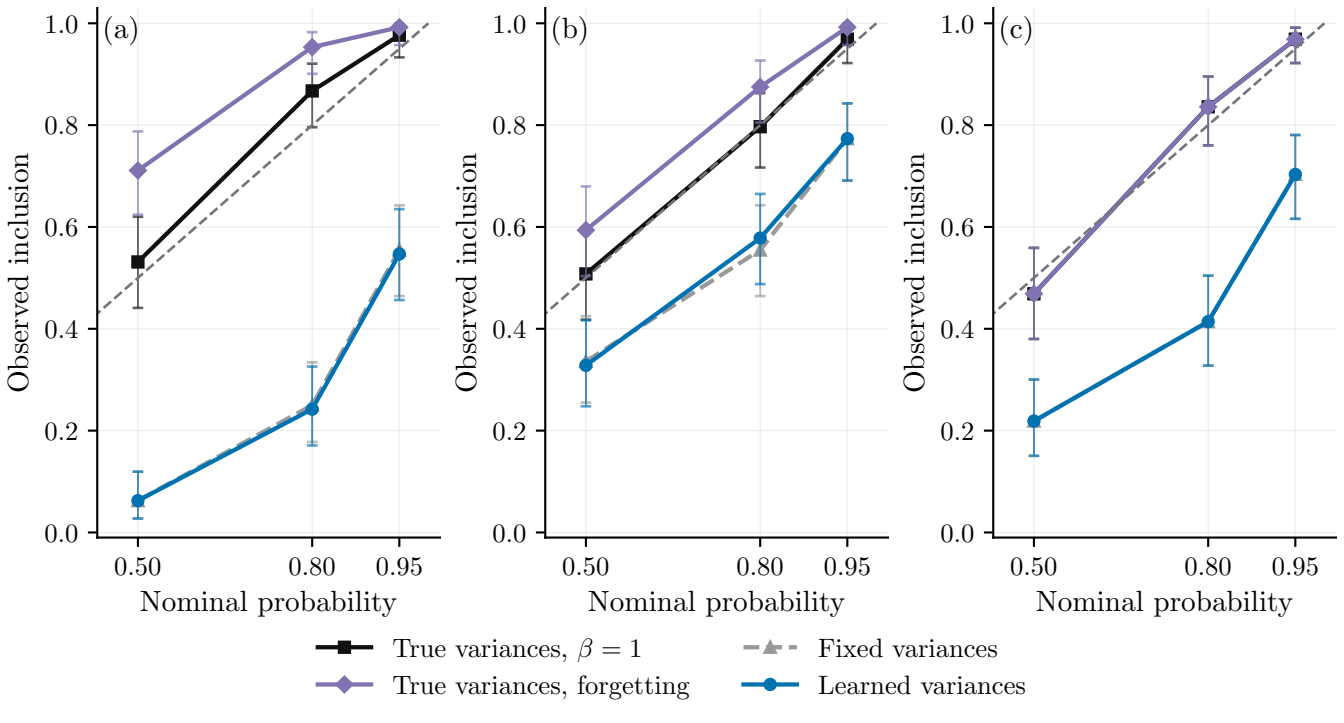

(a)
Noise variance σ^2
0.8
0.6
0.4
0.2
0.5
1.0
1.5
2.0
2.5
Time (s)
(b)
Signal variance τ^2
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.5
1.0
1.5
2.0
2.5
Time (s)
(c)
Output power / oracle power (dB)
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0.5
1.0
1.5
2.0
2.5
Time (s)
(d)
Predictive log score
0
−100
−200
−300
−400
0.5
1.0
1.5
2.0
2.5
Time (s)
True variances
Fixed variances
Learned variances
σ_t^2
τ_t^2

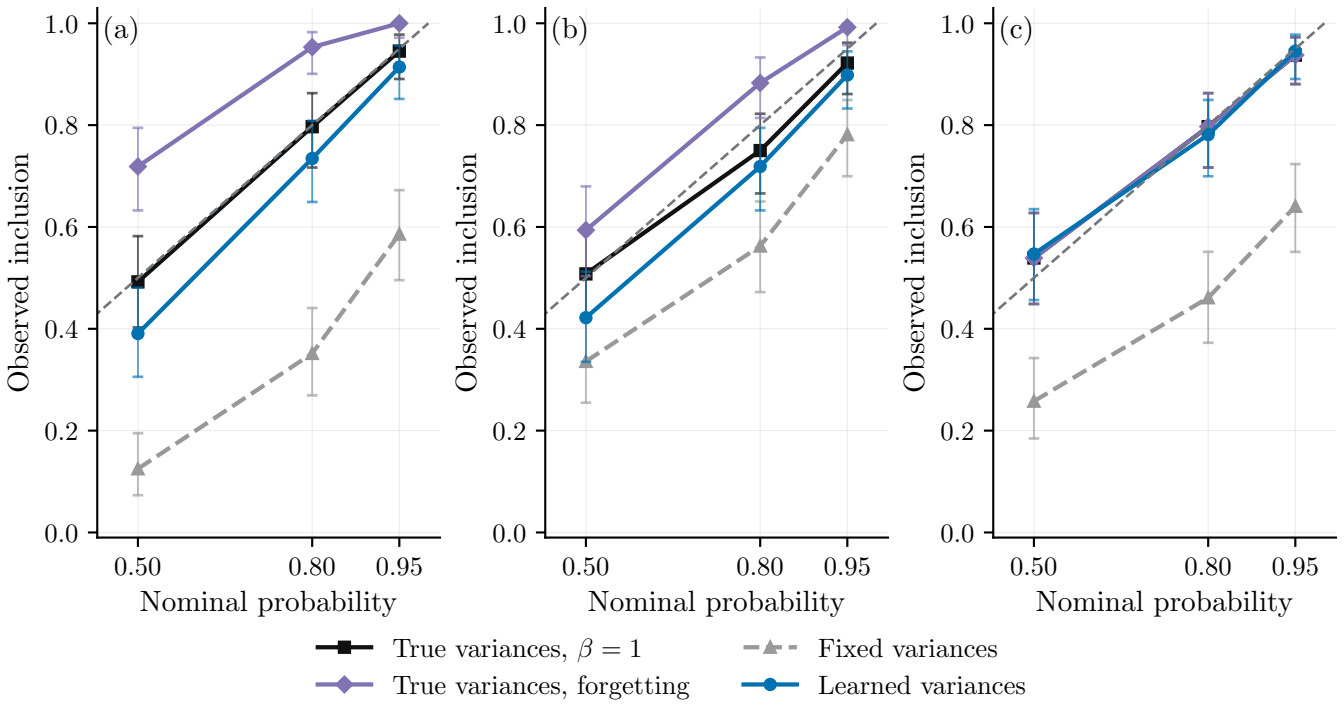

(a)
Noise variance σ^2
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.5
1.0
1.5
2.0
2.5
Time (s)
(b)
Signal variance τ^2
1.0
0.8
0.6
0.4
0.2
0.5
1.0
1.5
2.0
2.5
Time (s)
(c)
Output power / oracle power (dB)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0.5
1.0
1.5
2.0
2.5
Time (s)
(d)
Predictive log score
−20
−30
−40
−50
−60
−70
0.5
1.0
1.5
2.0
2.5
Time (s)
True variances
Learned variances
τ_t^2
Fixed variances
σ_t^2

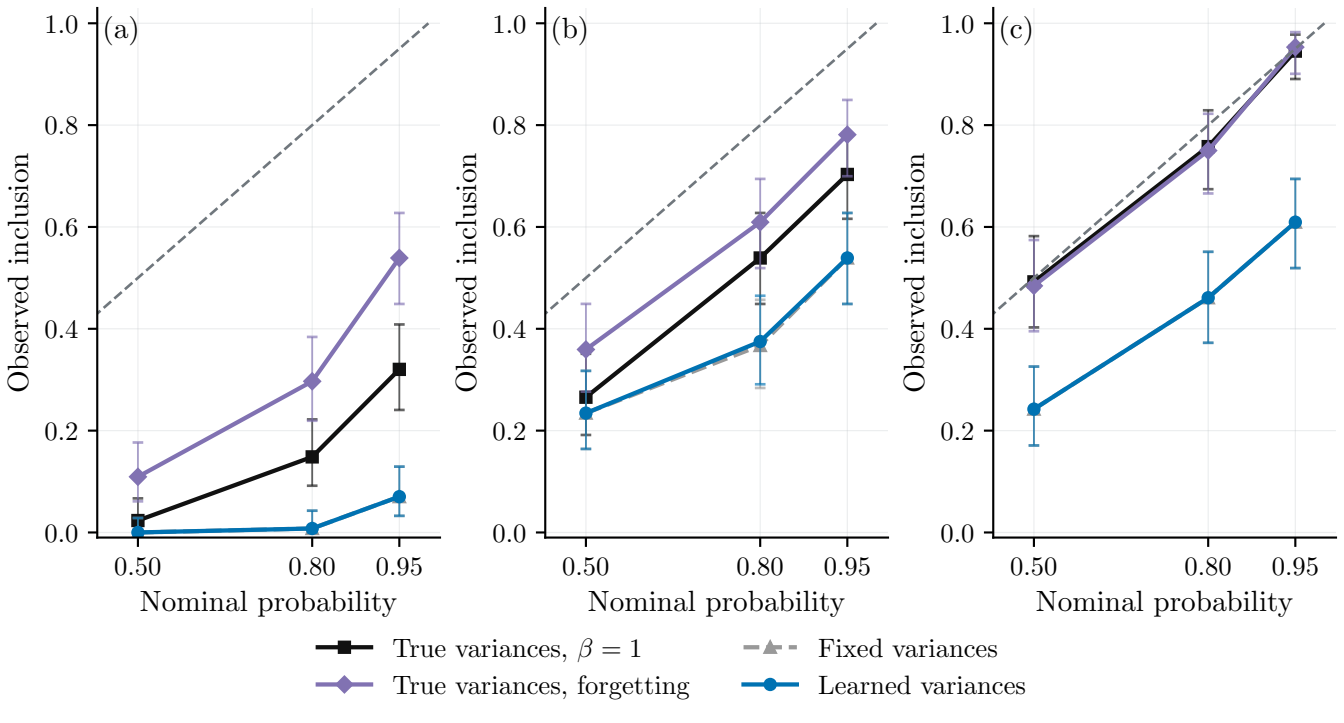

(a)
Noise variance σ^2
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.5
1.0
1.5
2.0
2.5
Time (s)
(b)
Signal variance τ^2
4
3
2
1
0
0.5
1.0
1.5
2.0
2.5
Time (s)
(c)
Output power / oracle power (dB)
2.5
2.0
1.5
1.0
0.5
0.0
0.5
1.0
1.5
2.0
2.5
Time (s)
(d)
Predictive log score
−20
−30
−40
−50
−60
−70
−80
0.5
1.0
1.5
2.0
2.5
Time (s)
True variances
Learned variances
τ_t^2
Fixed variances
σ_t^2

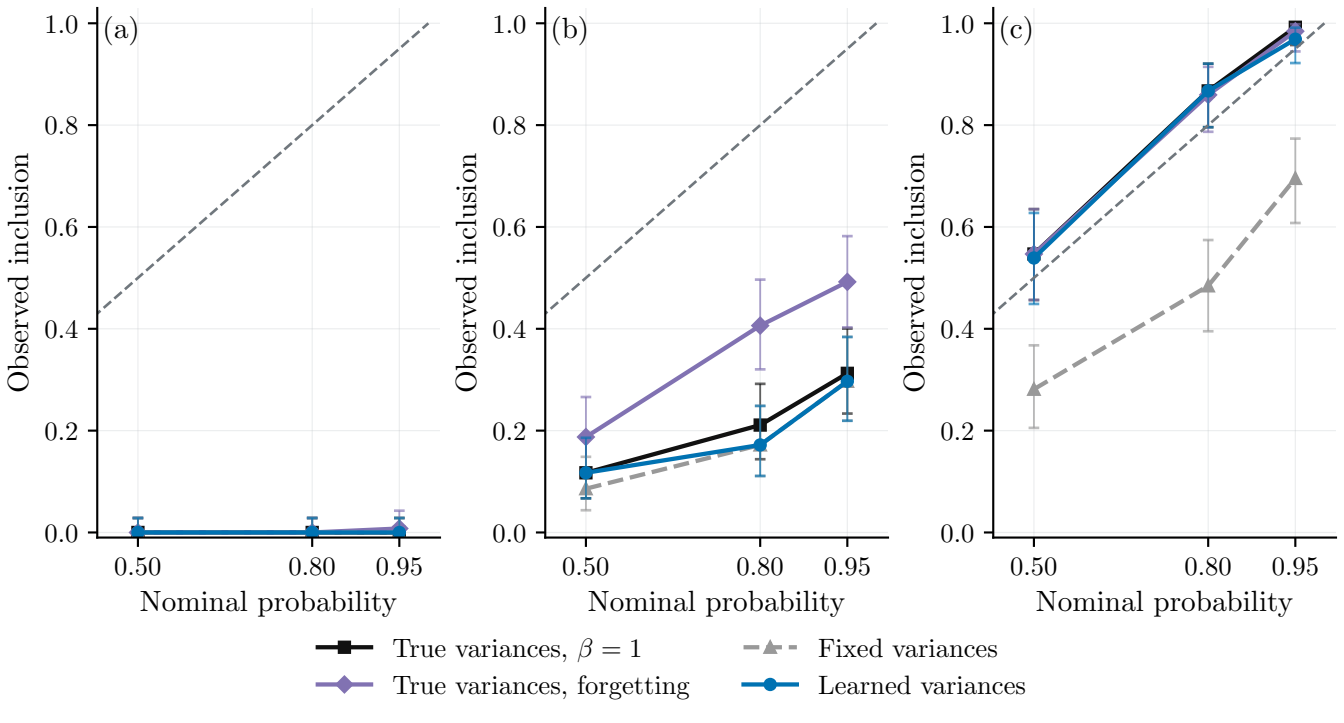

(a)
(b)
(c)
Observed inclusion
Nominal probability
0.0
0.2
0.4
0.6
0.8
1.0
0.50
0.80
0.95
True variances, $\beta = 1$
True variances, forgetting
Fixed variances
Learned variances

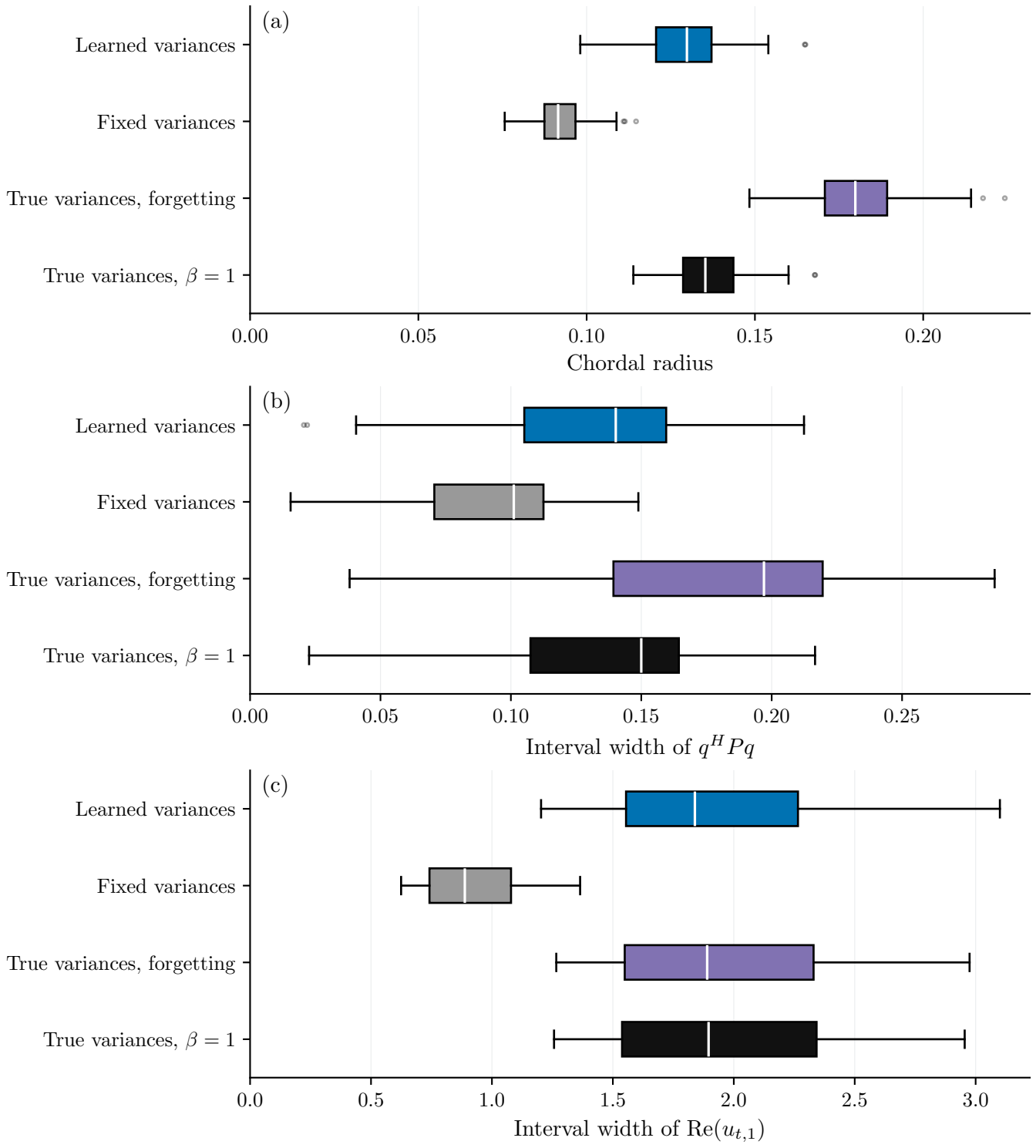

(a)
(b)
(c)
Observed inclusion
Nominal probability
0.0
0.2
0.4
0.6
0.8
1.0
0.50
0.80
0.95
True variances, $\beta = 1$
True variances, forgetting
Fixed variances
Learned variances

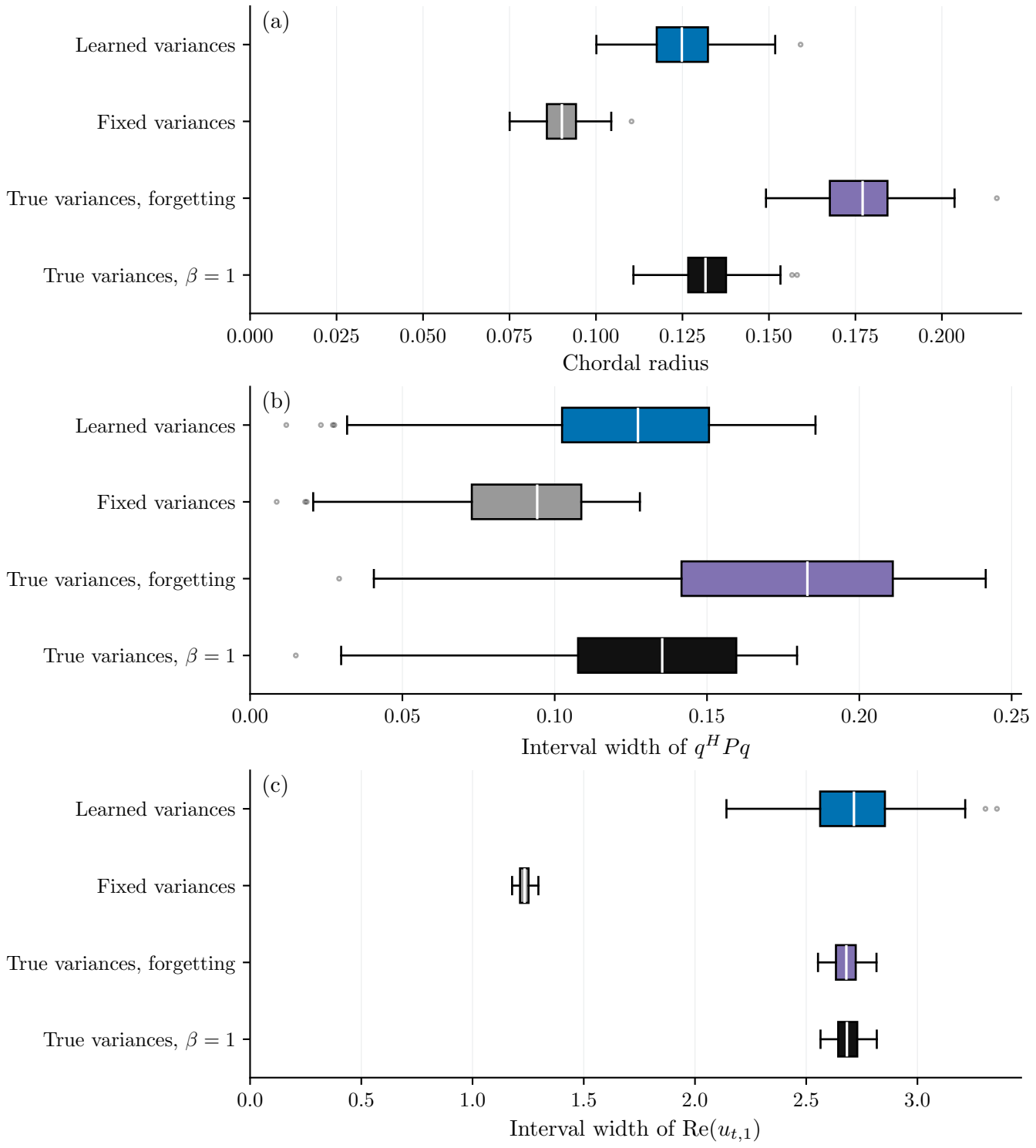

(a)
(b)
(c)
Observed inclusion
Nominal probability
0.0
0.2
0.4
0.6
0.8
1.0
0.50
0.80
0.95
True variances, $\beta = 1$
True variances, forgetting
Fixed variances
Learned variances

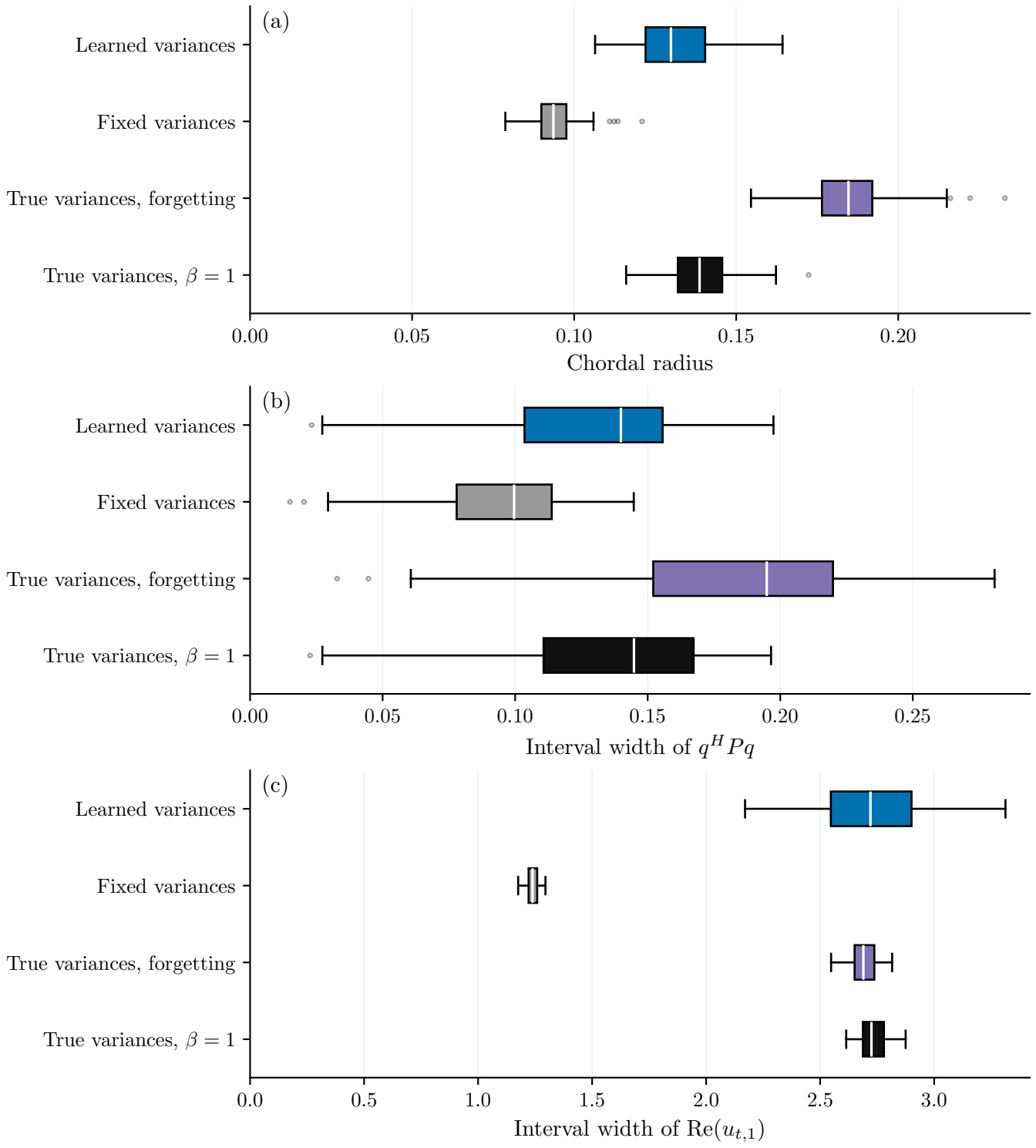

(a)
(b)
(c)
Observed inclusion
Nominal probability
0.0
0.2
0.4
0.6
0.8
1.0
0.50
0.80
0.95
True variances, $\beta = 1$
True variances, forgetting
Fixed variances
Learned variances

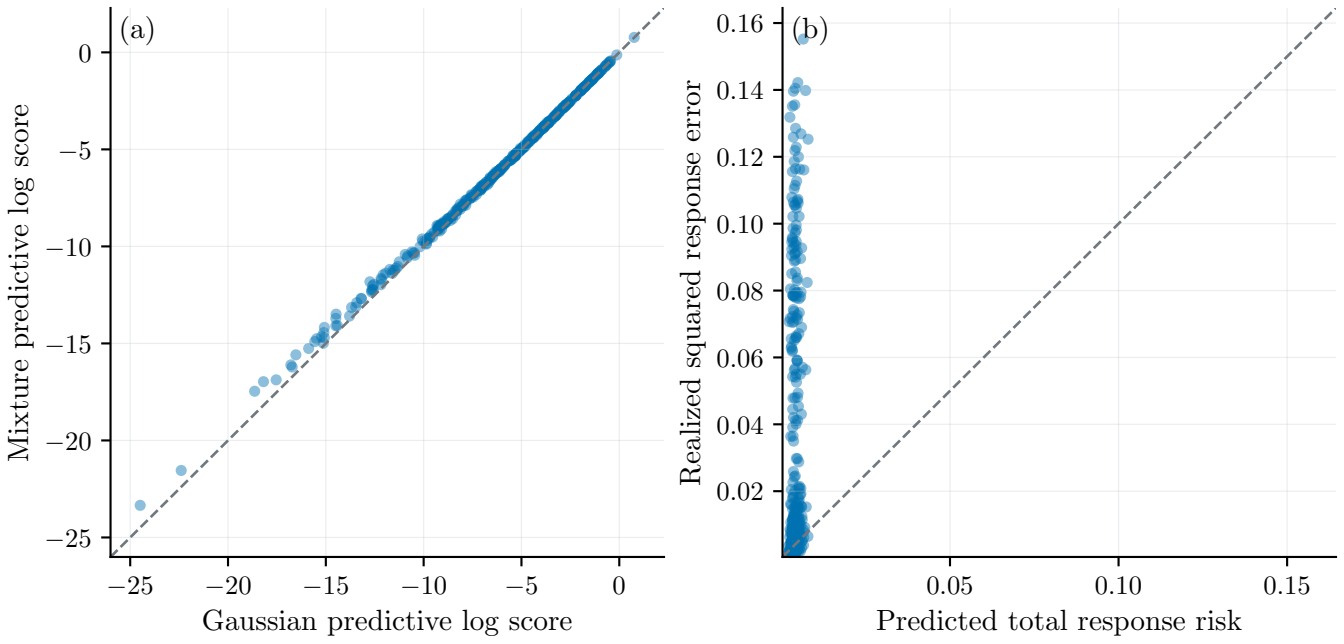

(a)
(b)
(c)
Observed inclusion
Nominal probability
0.0
0.2
0.4
0.6
0.8
1.0
0.50
0.80
0.95
True variances, $\beta = 1$
True variances, forgetting
Fixed variances
Learned variances

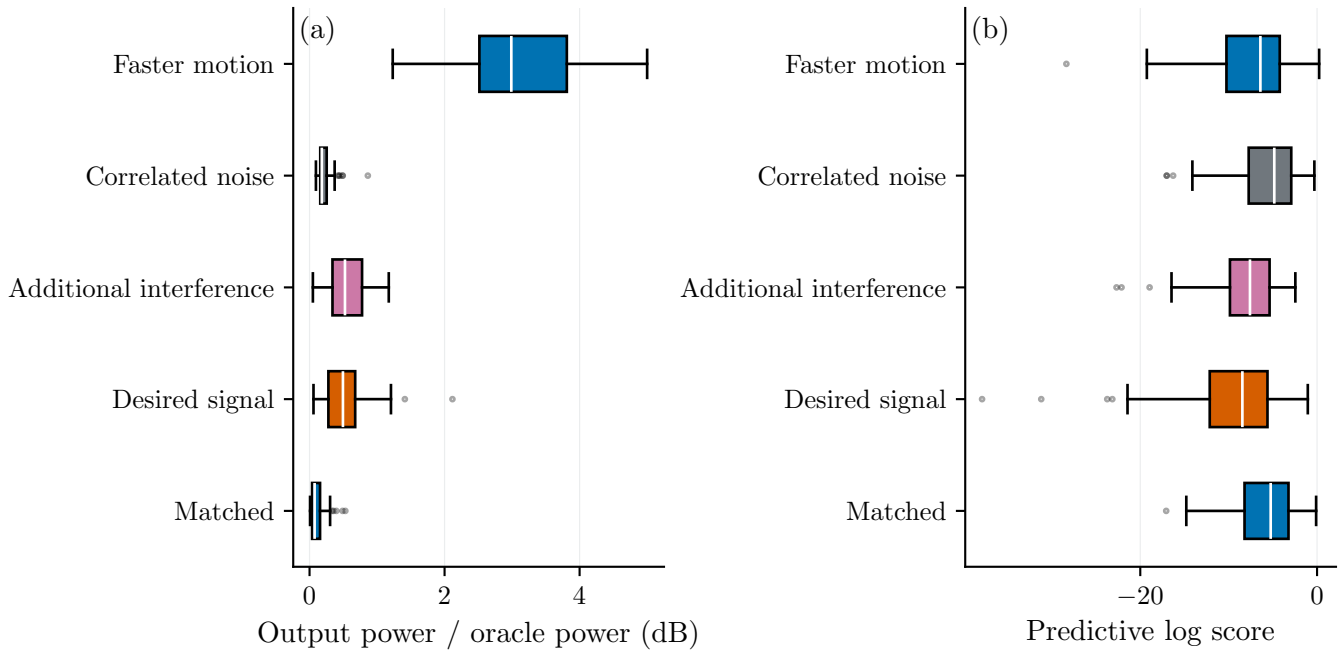

(a)
(b)
(c)
Observed inclusion
Nominal probability
0.0
0.2
0.4
0.6
0.8
1.0
0.50
0.80
0.95
True variances, $\beta = 1$
True variances, forgetting
Fixed variances
Learned variances

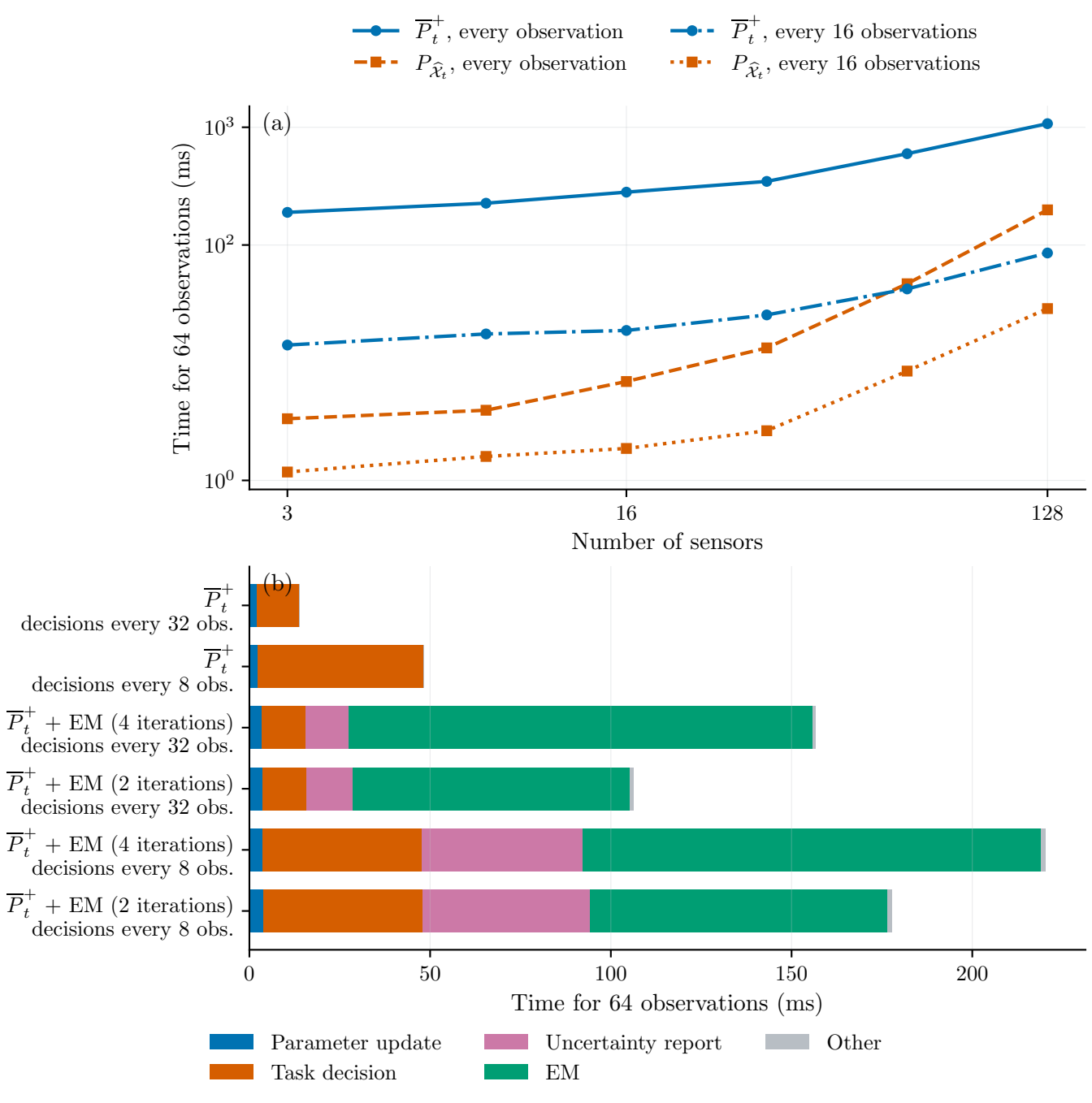

(a)
Learned variances
Fixed variances
True variances, forgetting
True variances, $\beta = 1$
0.00
0.05
0.10
0.15
0.20
Chordal radius
(b)
Learned variances
Fixed variances
True variances, forgetting
True variances, $\beta = 1$
0.00
0.05
0.10
0.15
0.20
0.25
Interval width of $q^H P q$
(c)
Learned variances
Fixed variances
True variances, forgetting
True variances, $\beta = 1$
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Interval width of $\mathrm{Re}(u_{t,1})$

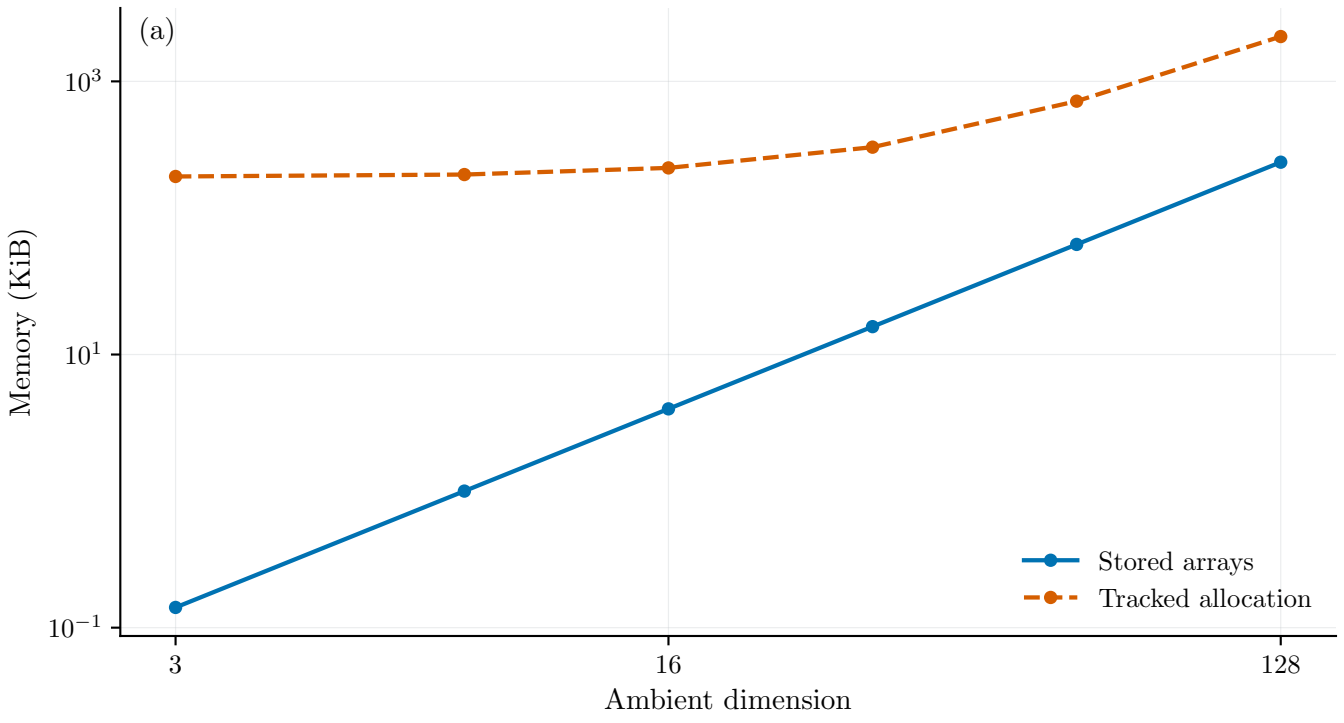

(a)
Learned variances
Fixed variances
True variances, forgetting
True variances, $\beta = 1$
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
Chordal radius
(b)
Learned variances
Fixed variances
True variances, forgetting
True variances, $\beta = 1$
0.00
0.05
0.10
0.15
0.20
0.25
Interval width of $q^H Pq$
(c)
Learned variances
Fixed variances
True variances, forgetting
True variances, $\beta = 1$
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Interval width of $\mathrm{Re}(u_{t,1})$

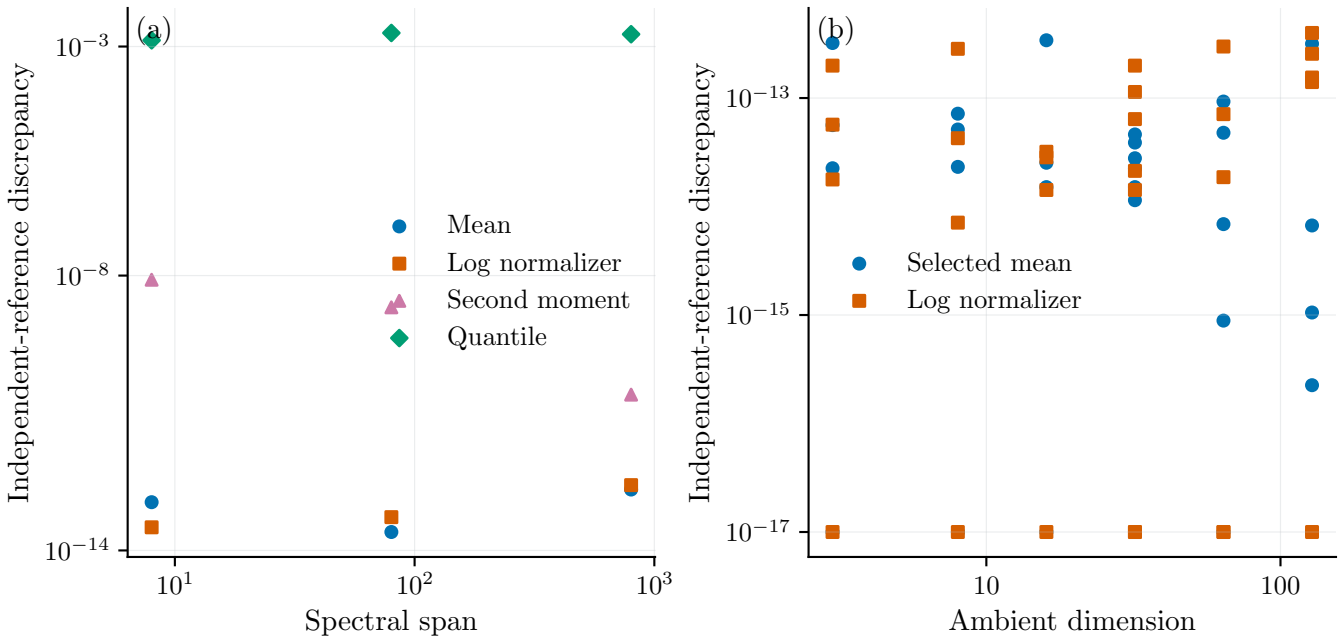

(a)
Learned variances
Fixed variances
True variances, forgetting
True variances, $\beta = 1$
0.00
0.05
0.10
0.15
0.20
Chordal radius
(b)
Learned variances
Fixed variances
True variances, forgetting
True variances, $\beta = 1$
0.00
0.05
0.10
0.15
0.20
0.25
Interval width of $q^H P q$
(c)
Learned variances
Fixed variances
True variances, forgetting
True variances, $\beta = 1$
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Interval width of $\mathrm{Re}(u_{t,1})$

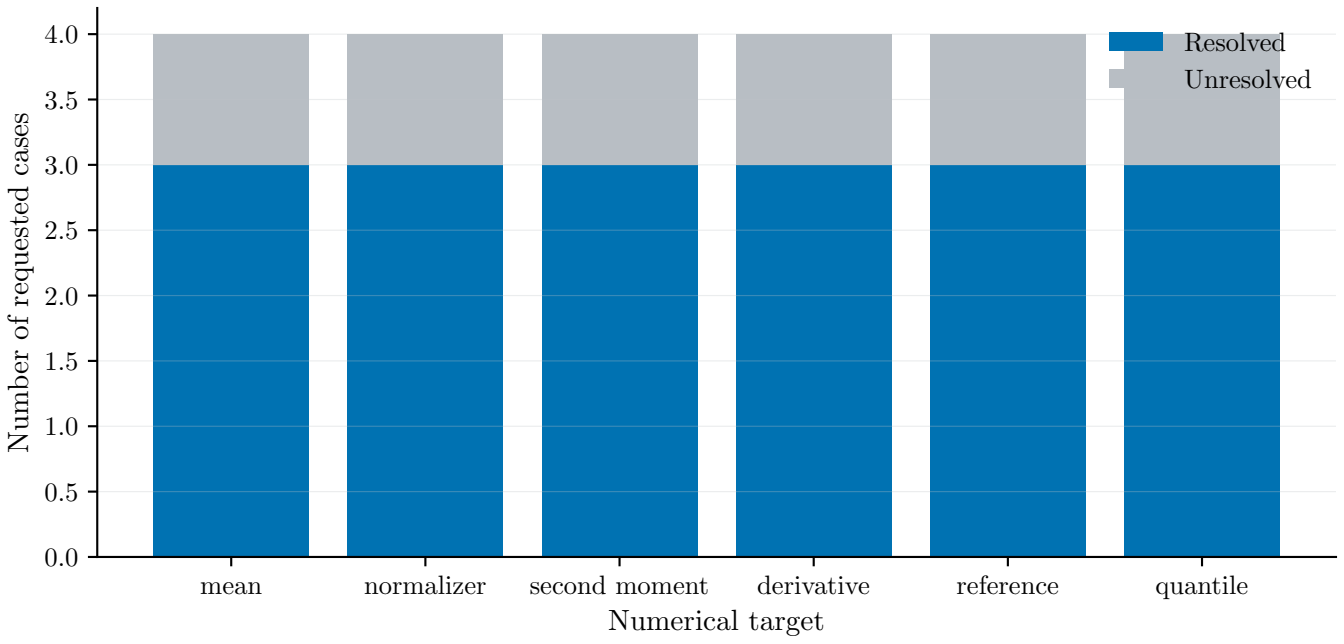

(a)
Mixture predictive log score
Gaussian predictive log score
0
−5
−10
−15
−20
−25
−25
−20
−15
−10
−5
0
(b)
Realized squared response error
Predicted total response risk
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.05
0.10
0.15

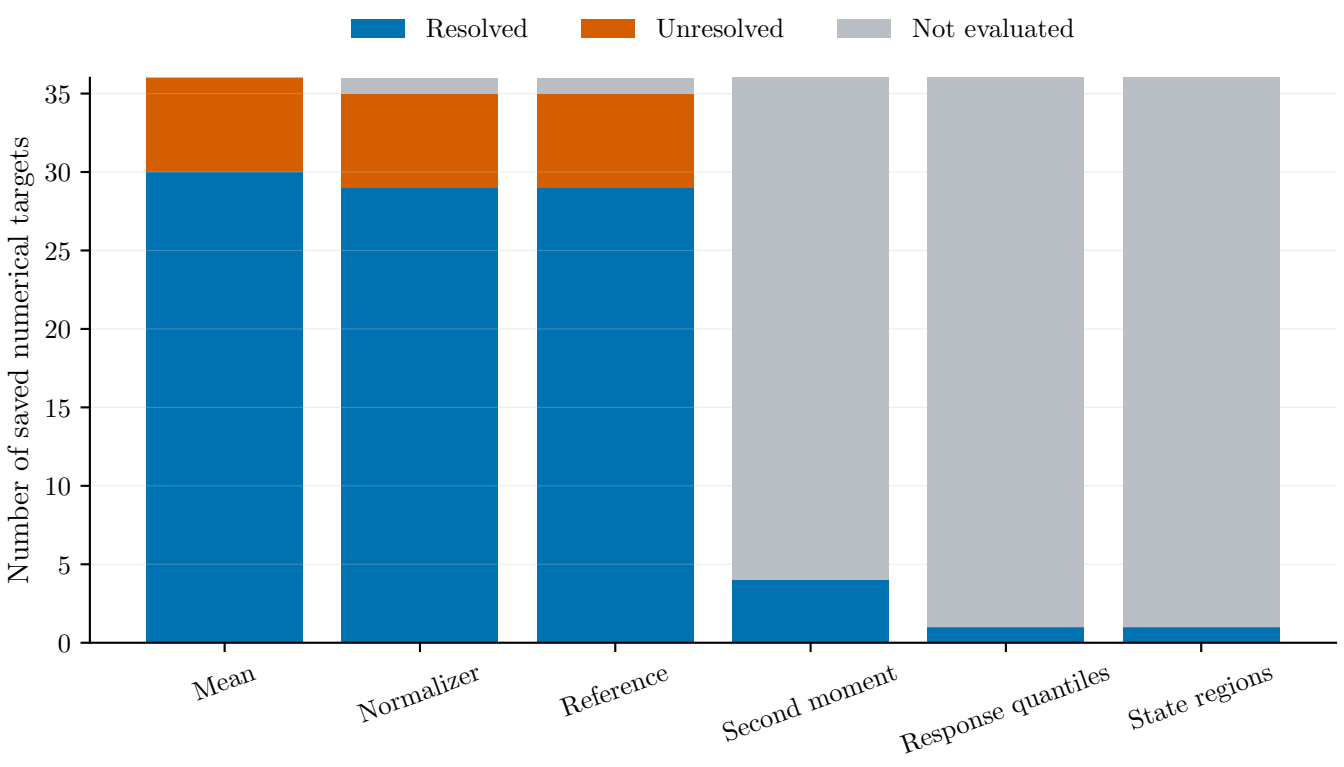

(a)
Faster motion
Correlated noise
Additional interference
Desired signal
Matched
0
2
4
Output power / oracle power (dB)
(b)
Faster motion
Correlated noise
Additional interference
Desired signal
Matched
−20
0
Predictive log score


$\overline{P}_t^+$, every observation
$\overline{P}_t^+$, every 16 observations
$P_{\widehat{\mathcal{X}}_t}$, every observation
$P_{\widehat{\mathcal{X}}_t}$, every 16 observations
(a)
Time for 64 observations (ms)
10^3
10^2
10^0
3
16
128
Number of sensors
(b)
$\overline{P}_t^+$ decisions every 32 obs.
$\overline{P}_t^+$ decisions every 8 obs.
$\overline{P}_t^+$ + EM (4 iterations) decisions every 32 obs.
$\overline{P}_t^+$ + EM (2 iterations) decisions every 32 obs.
$\overline{P}_t^+$ + EM (4 iterations) decisions every 8 obs.
$\overline{P}_t^+$ + EM (2 iterations) decisions every 8 obs.
0
50
100
150
200
Time for 64 observations (ms)
Parameter update
Task decision
Uncertainty report
EM
Other


(a)
10^3
10^1
10^{-1}
Memory (KiB)
Stored arrays
Tracked allocation
3
16
128
Ambient dimension


(a)
Independent-reference discrepancy
10^{-3}
10^{-8}
10^{-14}
10^{1}
10^{2}
10^{3}
Spectral span
Mean
Log normalizer
Second moment
Quantile
(b)
Independent-reference discrepancy
10^{-13}
10^{-15}
10^{-17}
10
100
Ambient dimension
Selected mean
Log normalizer


Resolved
Unresolved
Number of requested cases
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
mean
normalizer
second moment
derivative
reference
quantile
Numerical target


Resolved
Unresolved
Not evaluated
Number of saved numerical targets
0
5
10
15
20
25
30
35
Mean
Normalizer
Reference
Second moment
Response quantiles
State regions